CSE 4508 RDBMS
PROJECT PROPOSAL

Team DMA173 Presents

Project ClaritySync

Meet the team

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ClaritySync?

A modular, customizable business management system designed to handle inventory, sales, contacts, employees, banking, transactions, and warranty management.

Problem Domain

Small to medium-sized businesses struggle with managing multiple aspects of their operations using disconnected systems or manual processes, leading to inefficiency, errors, and poor decision-making.

Key Problems Addressed

- Fragmented customer & supplier data
- No real-time inventory or stock tracking
- Weak attendance tracking & inaccurate payslips
- No centralized financial/banking record
- No warranty or replacement tracking
- Manual tracking of debtors and creditors

Target Users

Retail businesses, wholesalers, service providers, and small manufacturing units that need an integrated solution but cannot afford enterprise-level ERP systems.

System Features & Modules

Toggleable Features

- Warranty tracking
- Employee attendance/payroll
- Banking module
- Serial number tracking

Contacts Module

Banking Module

Inventory Module

Sales Module

Employee Module

Warranty Module

Transactions Module

System Utilities

Role of Relational Database (RDB) in our project

- *Central Storage for All Business Data*
- *Ensures Data Consistency & Accuracy*
- *Supports Modular (Toggleable) Features*
- *Enables Fast Search, Filtering & Reporting*
- *Security, Access Control & Permissions*

Initial Schema Outline

SYSTEM_CONFIG

- config_id (PK)
- module_name (VARCHAR)
- is_enabled (BOOLEAN)
- setup_completed (BOOLEAN)
- business_name (VARCHAR)
- created_at (TIMESTAMP)

CONTACTS

- contact_id (PK)
- contact_type (ENUM: 'CUSTOMER', 'SUPPLIER', 'BOTH')
- name (VARCHAR)
- phone (VARCHAR)
- email (VARCHAR)
- address (TEXT)
- account_id (INT)
- created_at (TIMESTAMP)

PRODUCT

- product_id (PK)
- product_name (VARCHAR)
- description (TEXT)
- has_serial_number (BOOLEAN)
- warranty_enabled (BOOLEAN)
- warranty_period_days (INT)
- base_price (DECIMAL)
- created_at (TIMESTAMP)

INVENTORY

- inventory_id (PK)
- product_id (FK)
- serial_number (VARCHAR, NULLABLE)
- quantity (INT)
- purchase_price (DECIMAL)
- supplier_id (FK to CONTACT)
- purchase_date (DATE)
- status (ENUM: 'IN_STOCK', 'SOLD', 'WARRANTY_REPLACED')

Initial Schema Outline

EMPLOYEE

- employee_id (PK)
- name (VARCHAR)
- role (VARCHAR)
- designation (VARCHAR)
- basic_salary (DECIMAL)
- phone (VARCHAR)
- email (VARCHAR)
- join_date (DATE)
- is_active (BOOLEAN)

ATTENDANCE

- attendance_id (PK)
- employee_id (FK)
- date (DATE)
- check_in_time (TIME)
- check_out_time (TIME)
- status (ENUM: 'PRESENT', 'ABSENT', 'HALF_DAY', 'LEAVE')

BANKING_ACCOUNT

- account_id (PK)
- account_name (VARCHAR)
- account_type (ENUM: 'CASH', 'BANK', 'MOBILE_MONEY')
- account_number (VARCHAR, NULLABLE)
- current_balance (DECIMAL)
- created_at (TIMESTAMP)

TRANSACTION

- transaction_id (PK)
- transaction_type (ENUM: 'PAYMENT', 'RECEIVE', 'INVESTMENT', 'TRANSFER')
- amount (DECIMAL)
- from_account_id (FK, NULLABLE)
- to_account_id (FK, NULLABLE)
- contact_id (FK, NULLABLE)
- description (TEXT)
- transaction_date (TIMESTAMP)

Initial Schema Outline

SALES

- sale_id (PK)
- contact_id (FK, NULLABLE for walk-in customers)
- employee_id (FK to who made the sale)
- sale_date (TIMESTAMP)
- total_amount (DECIMAL)
- discount (DECIMAL)
- payment_method (VARCHAR)
- public_receipt_token (VARCHAR, UNIQUE)
- notes (TEXT)

SALE_ITEM

- sale_item_id (PK)
- sale_id (FK)
- product_id (FK)
- inventory_id (FK, for serialized items)
- quantity (INT)
- unit_price (DECIMAL)
- subtotal (DECIMAL)

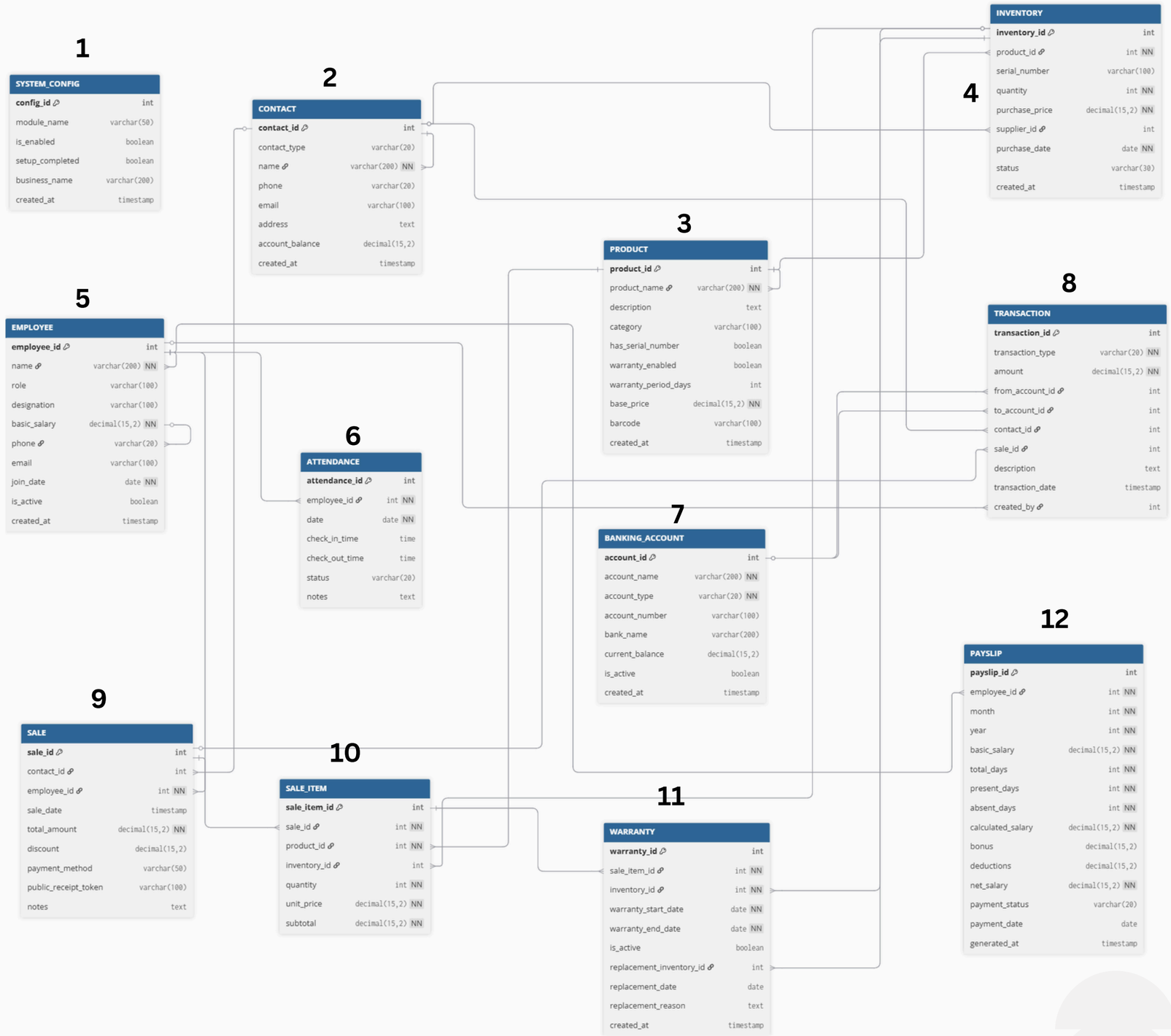
WARRANTY

- warranty_id (PK)
- sale_item_id (FK)
- inventory_id (FK original item)
- warranty_start_date (DATE)
- warranty_end_date (DATE)
- is_active (BOOLEAN)
- replacement_inventory_id (FK, NULLABLE)
- replacement_date (DATE, NULLABLE)
- replacement_reason (TEXT, NULLABLE)

PAYSLIP

- employee_id (PK)
- month (VARCHAR)
- year (INT)
- basic_salary (DEC)
- total_days (INT)
- present_days (INT)
- absent_days (INT)
- calculated_salary (DEC)
- bonus (DEC)
- deductions (DEC)
- net_salary (DEC)
- payment_date (DATE)

Draft ER Diagram



One-to-Many:

- (1) Contact → (9) Sale
- (1) Contact → (8) Trans.
- (3) Product→(10) Sale_item
- (3) Product→(4) Invent.
- (5) Emp. → (6) Attendance
- (5) Emp. → (12) Payslip
- (7) Banking_acc→(8) Trans.
- (9) Sale → (10) Sale_item
- (10) Sale_item → (11)Warr.

One-to-(zero or One):

- (4) Invent. → (11) Warr.

DatabaseComponents

Stored Procedures

- sp_CreateSale
- sp_ProcessWarrantyReplacement
- sp_GeneratePayslip
- sp_RecordTransaction
- sp_UpdateInventoryStock
- sp_TransferFunds
- sp_GetContactLedger

Triggers

- trg_AfterSaleInsert
- trg_UpdateContactBalance
- trg_UpdateBankingBalance
- trg_ValidateWarrantyDate
- trg_PreventDeleteActiveWarranty
- trg_LogInventoryChanges

Major SQL Queries

Business Intelligence Queries

- Daily/Monthly/Yearly Sales Reports with trends
- Top Selling Products Analysis
- Customer Purchase History and Lifetime Value
- Low Stock Alerts and Reorder Recommendations
- Employee Sales Performance Dashboard
- Profit Margin Analysis by Product/Category
- Outstanding Receivables and Payables Report
- Banking Account Reconciliation
- Warranty Claim Statistics
- Attendance and Payroll Summary

Complex Joins & Aggregations

- Multi-table joins for complete sale information with customer details
- Aggregate functions for inventory valuation
- Window functions for running balance calculations
- Subqueries for filtering products by warranty status
- CTEs for recursive employee hierarchy (if management levels added)

Technical Details

DBMS Choice: PostgreSQL
Frontend: React.js, HTML, CSS
Backend: Node.js(Express.js)
Authentication: bcrypt + JWT
Version Control: Git + GitHub
Deployment: Vercel + Free hosting
Testing: Postman + Jest

Team Roles

SADMAN

Database Responsibilities:

- System Config Module
- Banking & Transactions
- Auth/Security

Application Responsibilities:

- Backend Setup (Node/Express).

DIDHITI

Database Responsibilities:

- HR & Attendance
- Sales & Invoicing
- Analytics

Application Responsibilities:

- UI/UX Design (CSS/Tailwind).
- Dashboard Charts & PDF Generation (for Invoices/Payslips).

ANJIM

Database Responsibilities:

- Inventory & Products
- Contacts
- Warranty

Application Responsibilities:

- Frontend

Thank You

We are open for Q&A